

Data Sources



Drone

- Low-altitude video
- 3D mapping
- Post-disaster surveying



CCTV

- Real-time surveillance
- Object recognition



Connected Vehicles

- Traffic flow analysis
- Mobility patterns
- Street view imagery



Satellite

- Optical imagery
- SAR imagery
- Multispectral data



Hydrological Sensor

- Water level
- Flow velocity
- Water quality



Historical Data

- Disaster event record
- Response and recovery record
- Spatial distribution information



Social Media

- Situational awareness
- Information extraction
- Public sentiment analysis



Radar

- Weather monitoring
- Precipitation detection
- Storm tracking



Census

- Density distribution
- Building footprint
- Socioeconomic status

Events Spatial Distribution



Disaster Type

- Earthquake
- Explosion
- Hurricane
- Flooding
- Fire
- ...

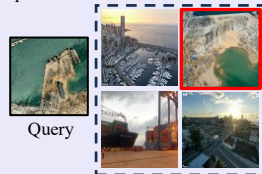
Alignment

Q1: Select all images depicting industrial zones with densely stacked container arrays.

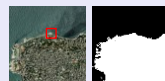


Q3: Given the remote sensing image and the corresponding DSM, determine the elevation at the marked point.

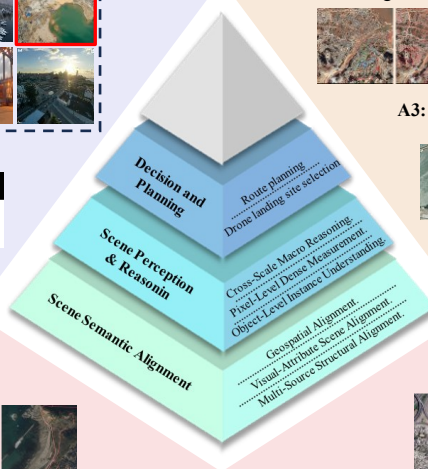
Q2: Please analyze the given satellite imagery and select the image that matches its spatial location from the options below.



Query



A3: 0-10m



Q3: An injured person needs to be transported from the current location to the designated destination for medical treatment. Based on road conditions and potential hazards, choose the safest and fastest route from the following candidates.



A3: Path A



A2: Region C

Decision & Planning

Q1: An industrial-grade hexacopter (approx. 1.2 meters in diameter, 15kg takeoff weight) is dispatched to deliver critical medical supplies to the disaster frontline. Can the drone safely land in this marked area?



A1: No, the area is filled with building debris

Q2: If the mission requires landing as close as possible to the top-most pier (where two ships are docked), which option best meets the safety requirements for a 1.2m diameter drone?

Perception & Reasoning

Q1: Please assess the damage level of the building within the bounding box by comparing the pre-disaster and post-disaster images.



A1: No visible structural damage; roof, walls, and footprint remain intact, with no clear signs of collapse, cracking, or deformation.

Q2: Analyze the damage condition of the specified road segment. Identify specific types of distress and assess the severity level.



A2: Massive landslide debris covering road, complete traffic interruption, high severity.

Q3: Based on the population distribution map, please estimate the number of people located within the bounding box shown in this region.



A3: 135 - 185

Q4: In pre-disaster images, the target building is a large, prominent oval stadium with a white roof and a rectangular green field in the center. Please assess the damage level of the building by comparing the pre-disaster and post-disaster images.



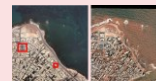
A4: Minor visible flooding; the building appears largely intact, with only limited water intrusion or slight signs of inundation, while the overall structure remains stable.

Q5: Given the image Ground Sample Distance (GSD) is 0.6 meters, please estimate the damaged building's floor area based on its outline in the image and output the result in square meters.



A5: 1000-3000 square meters

Q6: Given the image Ground Sample Distance (GSD) is 0.3 meters, please estimate the distance between the two objects based on their positions in the image and output the result in meters.



A6: 600-700 m